Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02 July 2026
Team ID	SWTID-2026-4978
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Technical Architecture Mappings:

S.No	Component	Description	Technology
1.	User Interface	Interactive analytics canvas layout for filtering and reviewing store shelf data dimensions.	Tableau Web UI / Embedded HTML5 & JavaScript
2.	Application Logic-1	Data abstraction, blending, and querying aggregation pipelines.	Tableau VizQL Engine
3.	Application Logic-2	Global parameter manipulation and cross-sheet relational action filtering logic.	Tableau Dashboard Action Manager
4.	Application Logic-3	In-database descriptive mathematical summaries and volume grouping logs.	Tableau Analytical Calculation Layer
5.	Database	Local text logs containing inventory configurations and store row IDs.	Flat CSV (Product Positioning.csv)
6.	Cloud Database	Cloud repository hosting workbook assets for operational review.	Tableau Cloud / Public Server Storage
7.	File Storage	Packed offline workbook configurations containing both layout logic and data layers.	Tableau Packaged Workbook (.twbx) System
8.	External API-1	Geocoding polygon mapping tool for rendering coordinates (if applicable).	None / Built-in Tableau OpenStreetMap map layer
9.	External API-2	Web client reference link out to regional marketplace competitor values.	None (Static file references used)
10.	Machine Learning Model	Native calculation matrices used to plot volume forecasting trend lines.	Tableau Native Linear Regression Model Engine
11.	Infrastructure (Server / Cloud)	Local development workspace and remote dashboard viewing environment.	Tableau Desktop (Local Windows) & Tableau Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web delivery scripts used to render workbook elements within dashboard containers.	Tableau JavaScript Embedding API v3
2.	Security Implementations	Row-level security parameters and local data storage isolation strategies.	Tableau Cloud User Permissions / Local File Encryption
3.	Scalability Architecture	High-speed columnar database extraction mechanics optimized for quick data parsing.	Tableau Hyper Extract Engine Architecture
4.	Availability	Cloud framework deployment ensuring high availability across web view browsers.	Tableau Server Cloud Architecture (99.9% Uptime)
5.	Performance	Caching calculation profiles and pre-compiling query contexts to ensure high response speeds.	Tableau Performance Optimization & Query Caching